Data Analysis

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
%matplotlib inline
warnings.simplefilter(action='ignore', category=FutureWarning)
```

In [2]:
```
train = pd.read_csv('emotion_detection.csv')
#train = pd.read_csv(r'C:\Users\EliteBook\Desktop\Lab02.Aqsa Khan.csv')
```

In [3]:
```
train.head()
```

Out[3]:

	word	limit	Age	category	disgust	surprise	neutral	anger	sad	
0	ability	0	16.0	5	0.004464	0.047832	0.000638	0.023597	0.013393	
1	able	1	23.0	6	0.000017	0.000182	0.000409	0.000176	0.000219	
2	abuse	1	25.0	7	0.000532	0.000177	0.000177	0.137363	0.001241	
3	academy	1	16.0	5	0.007143	0.021429	0.007143	0.007143	0.007143	
4	accept	0	33.0	6	0.008271	0.006767	0.000752	0.048872	0.018797	

In [4]:
```
train.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1104 entries, 0 to 1103
Data columns (total 12 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   word      1104 non-null   object 
 1   limit     1104 non-null   int64  
 2   Age       845 non-null    float64
 3   category  1104 non-null   int64  
 4   disgust   1104 non-null   float64
 5   surprise  1104 non-null   float64
 6   neutral   1104 non-null   float64
 7   anger     1104 non-null   float64
 8   sad       1104 non-null   float64
 9   happy     1104 non-null   float64
 10  fear      261 non-null    float64
 11  Sex       1104 non-null   object 
dtypes: float64(8), int64(2), object(2)
memory usage: 103.6+ KB
```

In [5]:
```
train.describe()
```

Out[5]:

	limit	Age	category	disgust	surprise	neutral
count	1104.000000	845.000000	1104.000000	1104.000000	1104.000000	1104.000000
mean	0.394928	31.111444	5.848732	0.012809	0.032376	0.008170
std	0.489057	17.168284	0.726233	0.015313	0.025154	0.010011
min	0.000000	0.750000	5.000000	0.000002	0.000003	0.000032
25%	0.000000	20.000000	5.000000	0.002528	0.015640	0.000992
50%	0.000000	29.000000	6.000000	0.007937	0.026907	0.003968
75%	1.000000	40.000000	6.000000	0.017857	0.041245	0.011905
max	1.000000	99.000000	7.000000	0.139456	0.135056	0.035714

Exploratory Data Analysis

In [6]:
```
#train.notnull()
train.isnull()
```

Out[6]:

	word	limit	Age	category	disgust	surprise	neutral	anger	sad	happy
0	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...
1099	False	False	False	False	False	False	False	False	False	False
1100	False	False	True	False	False	False	False	False	False	False
1101	False	False	False	False	False	False	False	False	False	False
1102	False	False	False	False	False	False	False	False	False	False
1103	False	False	True	False	False	False	False	False	False	False

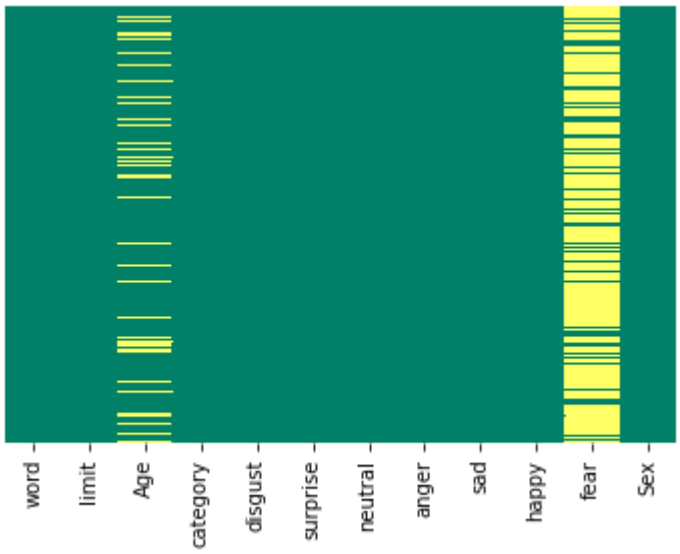
1104 rows × 12 columns

In [7]:
```
train.isnull().sum()
```

Out[7]:
```
word          0
limit         0
Age         259
category      0
disgust       0
surprise      0
neutral       0
anger         0
sad           0
happy         0
fear        843
Sex           0
dtype: int64
```

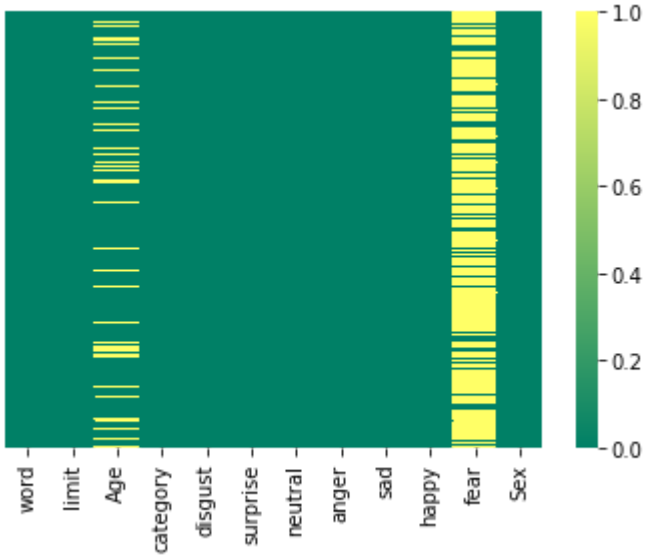
In [8]:
```
sns.heatmap(train.isnull(),yticklabels=False,cbar=False,cmap='summer')
```

Out[8]: <AxesSubplot:>



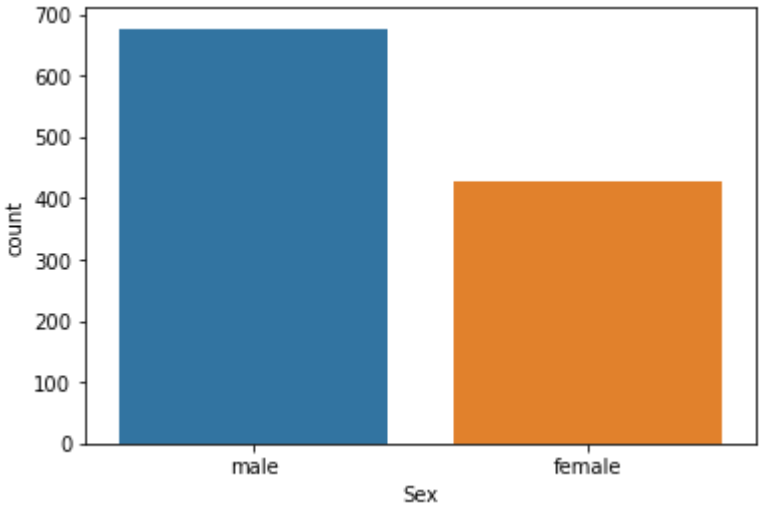
In [9]:
```
sns.heatmap(train.isnull(),yticklabels=False,cbar=True,cmap='summer')
```

Out[9]: <AxesSubplot:>



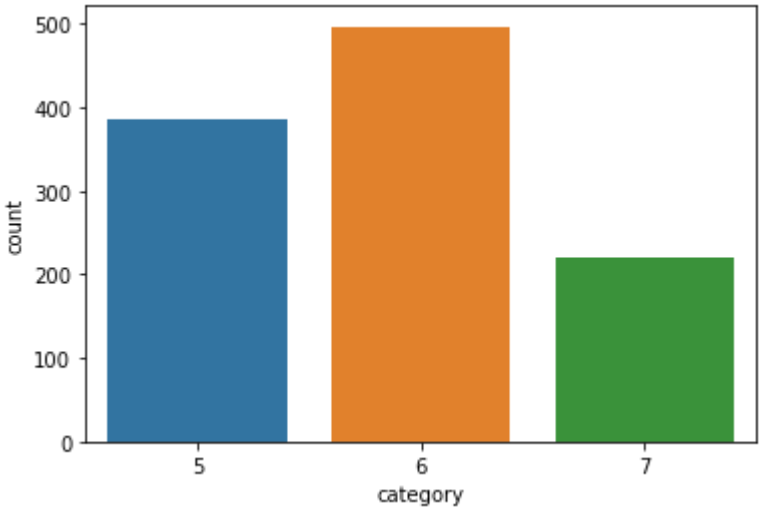
In [10]:
```
sns.countplot('Sex',data=train)
```

Out[10]: <AxesSubplot:xlabel='Sex', ylabel='count'>



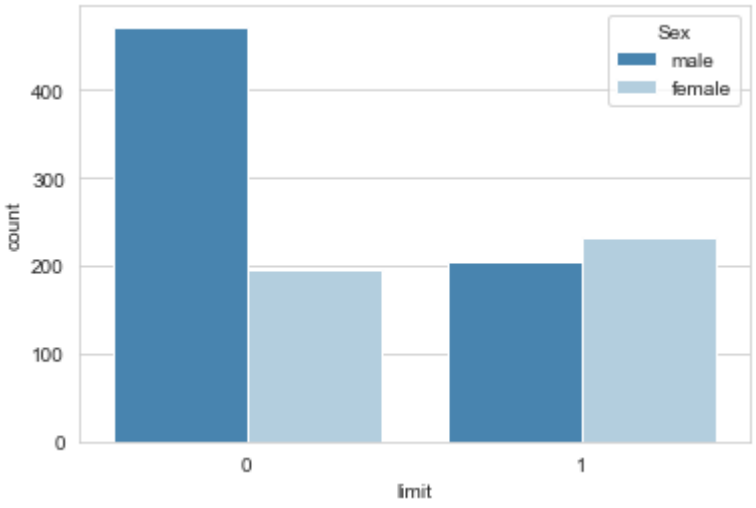
In [11]:
```
sns.countplot('category',data=train)
```

Out[11]: <AxesSubplot:xlabel='category', ylabel='count'>



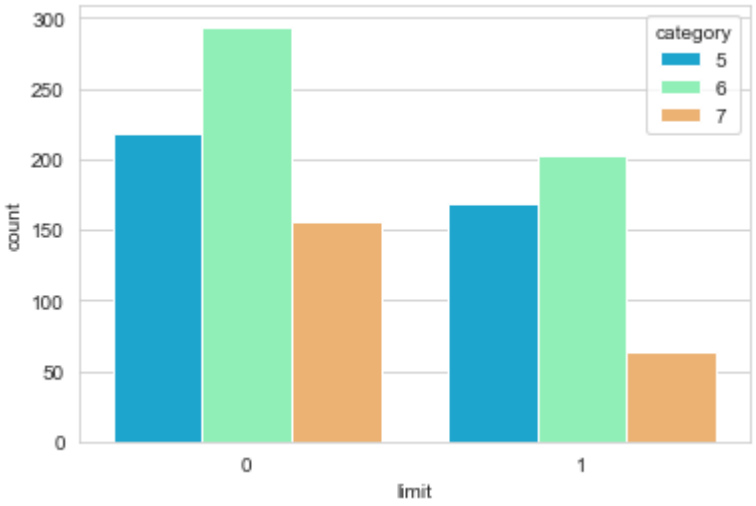
In [12]:
```
sns.set_style('whitegrid')
sns.countplot(x='limit',hue='Sex',data=train,palette='Blues_r')
```

Out[12]: <AxesSubplot:xlabel='limit', ylabel='count'>



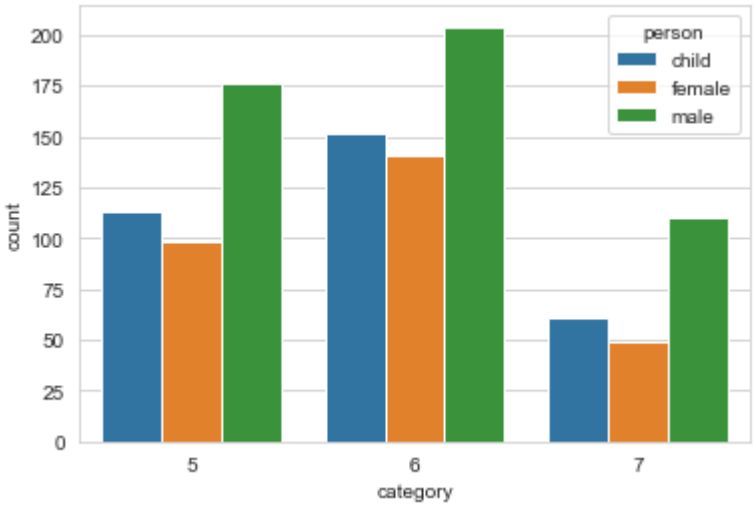
In [13]:
```
sns.set_style('whitegrid')
sns.countplot(x='limit',hue='category',data=train,palette='rainbow')
```

Out[13]: <AxesSubplot:xlabel='limit', ylabel='count'>



In [15]:
```
def number_children(category):
    Age, Sex= category
    if Age < 25:
        return 'child'
    else:
        return Sex
train['person'] = train[['Age','Sex']].apply(number_children, axis=1)
sns.countplot('category', data=train, hue='person')
```

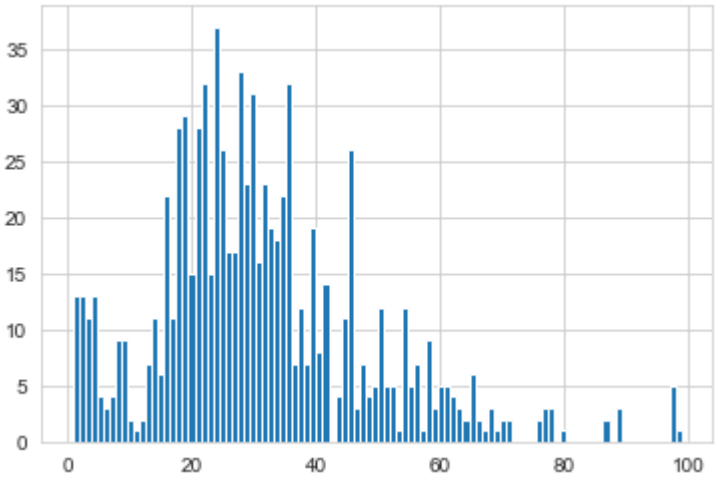
Out[15]: <AxesSubplot:xlabel='category', ylabel='count'>



Histogram

In [16]:
```
train['Age'].hist(bins=100)
```

Out[16]: <AxesSubplot:>



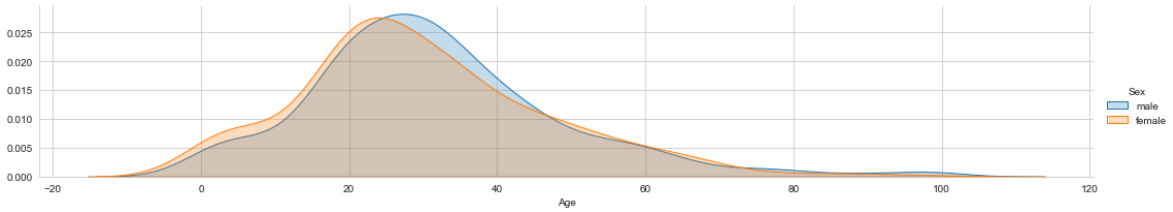
In [17]:
```
#Comparison of male, female, and children
train['person'].value_counts()
```

Out[17]:
```
male      490
child     326
female    288
Name: person, dtype: int64
```

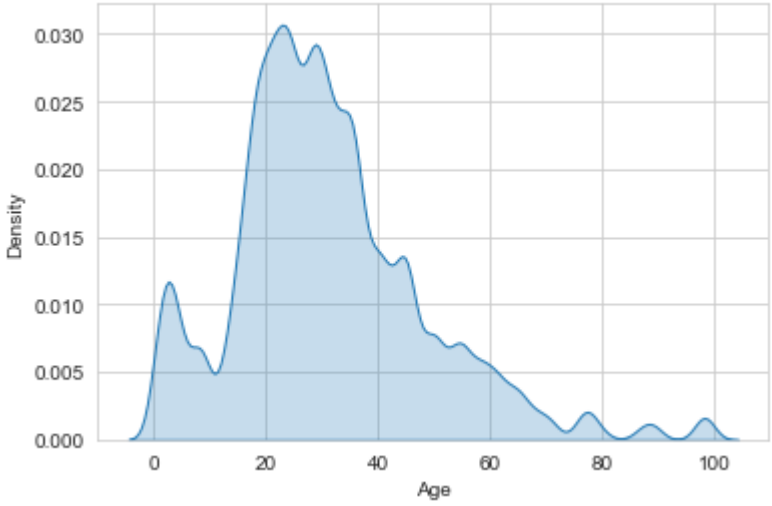
KDE Plot (Smooth Distribution)

In [18]:
```
fig = sns.FacetGrid(train, hue="Sex",aspect=5)
fig.map(sns.kdeplot,'Age',shade= True)
fig.add_legend()
```

Out[18]:
```
<seaborn.axisgrid.FacetGrid at 0x2cde8d22520>
```



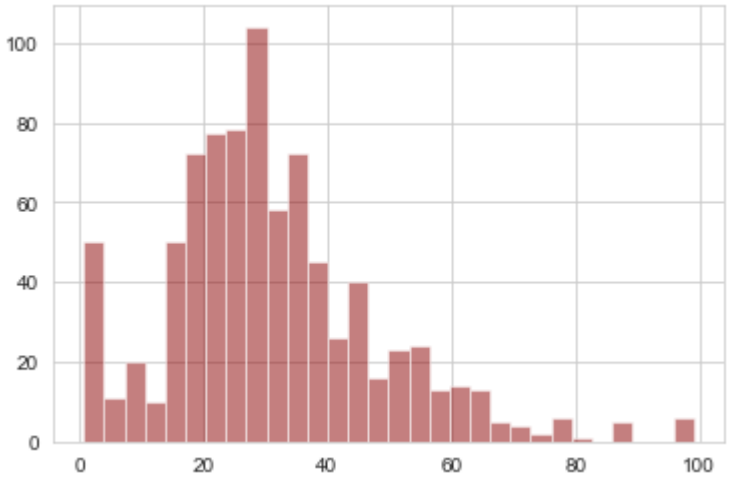
In [20]:
```
#bandwidth effect can be seen here with increasing or decreasing the valu
sns.kdeplot(train['Age'], shade=True, bw=0.1)
plt.show()
```



Distribution Plot

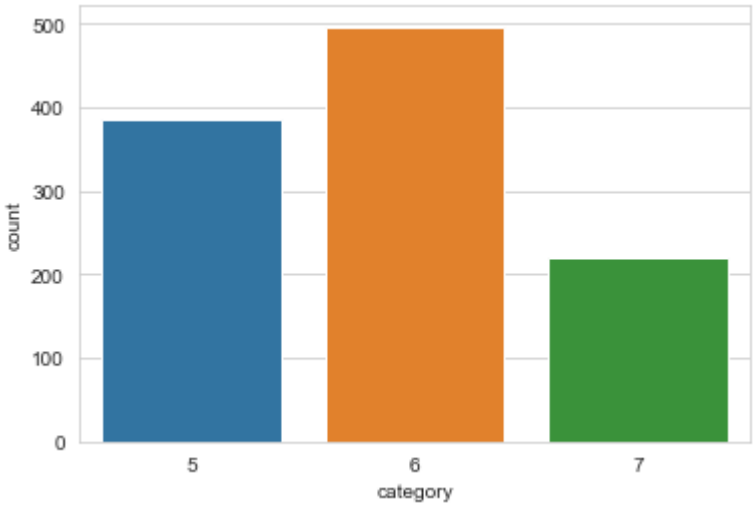
In [22]:
```
train['Age'].hist(bins=30,color='darkred',alpha=0.5)
```

Out[22]: <AxesSubplot:>



In [23]:
```
#determining the number of siblings and spouse. 6 means no siblings, 5 me
sns.countplot(x='category',data=train)
```

Out[23]: <AxesSubplot:xlabel='category', ylabel='count'>

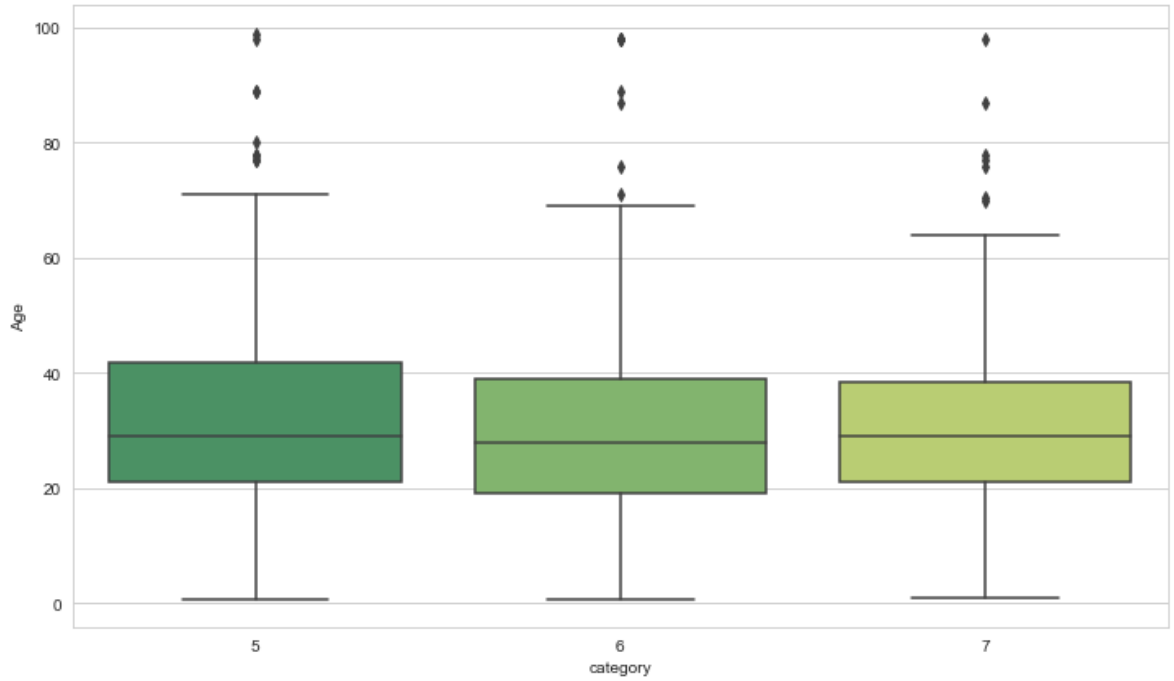


Data Cleaning

In [22]:
```
#Box Plot is the visual representation of the depicting groups of numeric
#Boxplot is also used to detect the outlier in data set.
#box plot is consisting of :Minimum, First Quartile or 25%, Median (Secon

plt.figure(figsize=(12, 7))
sns.boxplot(x='category',y='Age',data=train,palette='summer')
```

Out[22]: <AxesSubplot:xlabel='category', ylabel='Age'>



In [23]:
```
train['Age'].mean()
```

Out[23]: 31.111443786982246

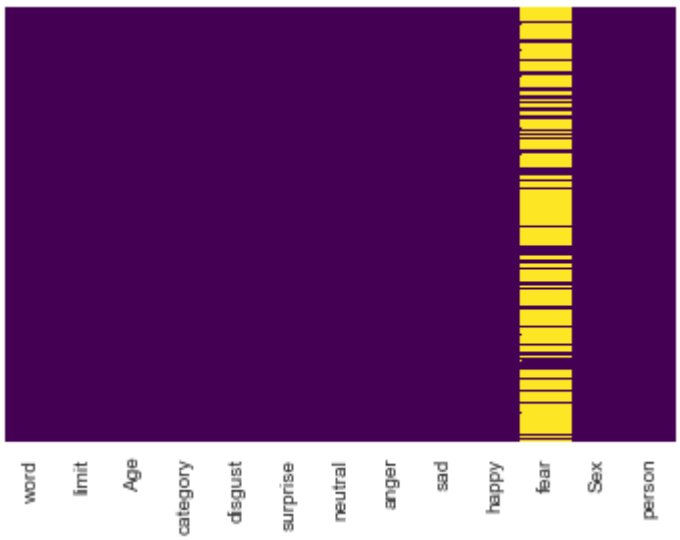
In [26]:
```
#Simply take replace the missing age with the mean age
train['Age'] = train['Age'].fillna(train['Age'].mean())
```

In [27]:
```
train['Age'].mean()
```

Out[27]: 29.443088768115942

In [28]:
```
sns.heatmap(train.isnull(),yticklabels=False,cbar=False,cmap='viridis')
```

Out[28]: <AxesSubplot:>



In [29]:
```
train.head()
```

Out[29]:

	word	limit	Age	category	disgust	surprise	neutral	anger	sad	
0	ability	0	16.0	5	0.004464	0.047832	0.000638	0.023597	0.013393	
1	able	1	23.0	6	0.000017	0.000182	0.000409	0.000176	0.000219	
2	abuse	1	25.0	7	0.000532	0.000177	0.000177	0.137363	0.001241	
3	academy	1	16.0	5	0.007143	0.021429	0.007143	0.007143	0.007143	
4	accept	0	33.0	6	0.008271	0.006767	0.000752	0.048872	0.018797	

In [30]:
```
train.drop('fear',axis=1,inplace=True)
```

In [31]:
```
train.head()
```

Out[31]:

	word	limit	Age	category	disgust	surprise	neutral	anger	sad	
0	ability	0	16.0	5	0.004464	0.047832	0.000638	0.023597	0.013393	
1	able	1	23.0	6	0.000017	0.000182	0.000409	0.000176	0.000219	
2	abuse	1	25.0	7	0.000532	0.000177	0.000177	0.137363	0.001241	
3	academy	1	16.0	5	0.007143	0.021429	0.007143	0.007143	0.007143	
4	accept	0	33.0	6	0.008271	0.006767	0.000752	0.048872	0.018797	

In [33]:
```
train['anger'].mode()
```

Out[33]:
```
0    0.035714
dtype: float64
```

Lab Tasks

1. You people needs to explore dataset from kaggle

2. Download the dataset and apply all the steps that we have done.

In [ ]:
```

```